

HealthConnect Clinic

Experience Lab Internship Programme

Week 7 Assignment

Data Science Track

Model Testing, Error Analysis & Refinement

SOGA Para

Data Science Track

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September 20, 2026

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Executive Summary

This report presents the work carried out during Week 7 of the AnalystLab Africa Experience Lab programme, as part of the HealthConnect Clinic project. The main objective of this week was to test, refine and validate the candidate model developed during Week 6.

The main achievements of this week include:

- **6 systematic tests** carried out on the candidate model
- **10-fold cross-validation** to assess robustness
- **Generalisation test** on unseen data
- **Stability test** across 5 random seeds
- **Decision threshold optimisation**
- **Segment analysis** (age groups)
- **Overfitting detection**
- **HC-POD collaboration** with the Data Analytics track

The most important result of this week is the demonstration that the model is **stable** (standard deviation = 0.0186), **free of overfitting** (Train/Test gap = 0.0014) and **homogeneous** across all segments ($F1 > 0.60$). Three out of six tests failed, but the gaps are very small (between -0.005 and -0.016).

1 Week 6 → Week 7 Transition

1.1 Main Output of Week 6

The main deliverable of Week 6 was a **candidate binary classification model** using logistic regression. This model aimed to predict missed appointments (No-Show) at the HealthConnect clinic.

Table 1: Candidate Model Metrics - Week 6

Metric	Score
Accuracy	62.13%
Precision	62.07%
Recall	66.80%
F1-Score	64.35%
ROC-AUC	66.02%
Optimal threshold	0.30
F1 at optimal threshold	69.45%

1.2 Most Important Component

The most important component identified in Week 6 is the **No-Show prediction model** with the following characteristics:

- Model: Logistic Regression
- Features: 16 variables
- Optimal threshold: 0.30
- 5-fold cross-validation: 0.6403 ± 0.0102

1.3 Main Limitation Identified

Table 2: Limitations Identified in Week 6

Limitation	Description
False Negatives	161 undetected cases
External validation	No test on unseen data
Robustness	No in-depth stability test

1.4 Component to Test

The component to test in Week 7 is the **No-Show prediction model** with the following objectives:

- Test robustness (10-fold cross-validation)

- Test generalisation (unseen data)
- Test stability (random seeds)
- Optimise the decision threshold
- Analyse performance by segment
- Detect overfitting

1.5 Relevant Tracks

Table 3: Relevant Tracks for Week 7

Track	Relevance
Data Analytics	Validation of KPIs and insights
ML Engineering	Testing of the integration pipeline

2 Testing Approach

2.1 Test Objectives

Table 4: Test Objectives

Test	Objective	Criterion
Test 1	Robustness	$F1 > 0.65$
Test 2	Generalisation	$F1 > 0.65$
Test 3	Stability	Standard deviation < 0.02
Test 4	Threshold optimisation	$F1 > 0.70$
Test 5	Segment performance	$F1 > 0.60$
Test 6	Overfitting	Gap < 0.05

2.2 Methodology

1. **10-fold cross-validation:** Assess the model's robustness across 10 folds
2. **Generalisation test:** Test on a different split (`random_state=123`)
3. **Stability test:** Test across 5 different random seeds
4. **Threshold optimisation:** Test different thresholds from 0.1 to 0.9
5. **Segment performance:** Analyse by age group
6. **Overfitting detection:** Compare Train vs Test

3 Test Results

3.1 Test 1: 10-fold Cross-Validation

Table 5: 10-fold Cross-Validation Results

Model	Average F1	Standard deviation
Baseline (LR)	0.6448	± 0.0204
Random Forest	0.6221	± 0.0247

Interpretation: The average F1 (0.6448) is very close to the threshold (0.65). The standard deviation (0.0204) is low, indicating good stability. The Random Forest is less performant than the baseline.

3.2 Test 2: Generalisation (Unseen Data)

Table 6: Generalisation Test Results

Metric	Result	Criterion
F1-Score	0.6336	> 0.65

Interpretation: The model generalises slightly less well on a new split (0.6336 vs 0.6435). The difference is small (-0.01), indicating good consistency.

3.3 Test 3: Stability (Random Seeds)

Table 7: Stability Test Results

Metric	Result	Criterion
Mean	0.6407	-
Standard deviation	0.0186	< 0.02

Interpretation: The model is **stable** (standard deviation < 0.02). Good reproducibility.

3.4 Test 4: Threshold Optimisation

Table 8: Threshold Optimisation Results

Metric	Result	Criterion
Optimal threshold	0.30	-
Optimal F1-Score	0.6945	> 0.70

Interpretation: The optimal threshold (0.30) improves the F1 by **+5.1 points** (0.6435 $\rightarrow$ 0.6945). Very close to the 0.70 criterion.

3.5 Test 5: Segment Performance

Table 9: Performance by Age Group

Age group	F1	Patients	Criterion
18-34	0.6549	267	> 0.60
35-49	0.6589	231	> 0.60
50-64	0.6412	237	> 0.60
65+	0.6108	213	> 0.60

Interpretation: Performances are **homogeneous** across all segments. The 65+ group has the lowest F1 (0.6108).

3.6 Test 6: Overfitting Detection

Table 10: Train vs Test Comparison

Metric	Result	Criterion
F1 Train	0.6449	-
F1 Test	0.6435	-
Gap	0.0014	< 0.05

Interpretation: **No overfitting** (very small gap). The model generalises well.

4 Test Summary

Table 11: Summary of Tests

Test	Result	Criterion	Status	Gap
10-fold cross-validation	0.6448	> 0.65	FAIL (close)	-0.005
Generalisation	0.6336	> 0.65	FAIL (close)	-0.016
Stability	0.0186	< 0.02	PASS	-
Threshold optimisation	0.6945	> 0.70	FAIL (close)	-0.005
Segment performance	0.61-0.66	> 0.60	PASS	-
Overfitting	0.0014	< 0.05	PASS	-

Summary: 3 PASS / 3 FAIL

4.1 Interpretation

The 3 failed tests are **very close** to the objectives:

- Cross-validation: 0.6448 vs 0.65 (gap of -0.005)
- Generalisation: 0.6336 vs 0.65 (gap of -0.016)
- Threshold optimisation: 0.6945 vs 0.70 (gap of -0.005)

These gaps are **minor** and the model remains usable.

The 3 passed tests confirm the robustness of the model:

- Stability: standard deviation < 0.02
- Segment performance: all > 0.60
- Overfitting: gap < 0.05

5 HC-POD Cross-Track Collaboration

5.1 Collaborated Track

The **Data Analytics** track was identified as the partner for this collaboration.

5.2 Project Dependency

Table 12: Dependency with Data Analytics

Aspect	Detail
Dependency	Validated KPIs inform modelling decisions
Justification	Insights guide feature selection

5.3 Testing Activity

1. Cross-validation of KPIs with model predictions
2. Verification of segment consistency
3. Analysis of gaps between KPIs and predictions

5.4 Identified Problem

Gap between model predictions and KPIs on certain segments.

5.5 Refinement

- Adjustment of the decision threshold ($0.50 \rightarrow 0.30$)
- Validation of features by segment

5.6 Re-test

F1-Score after refinement: 0.6945 (vs 0.6435 before)

5.7 Resulting Change

Table 13: Changes following the Collaboration

Change	Description
Improved consistency	Better match with KPIs
Better interpretability	Features validated by segment
Performance gain	+5.1 F1 points

5.8 Evidence

- Exchange file (Google Drive)
- Screenshot of messages
- GitHub commit

6 Validation and Recommendations

6.1 Model Validation

Table 14: Validation Criteria

Criterion	Objective	Achieved	Status
Cross-validation	> 0.65	0.6448	(close)
Generalisation	> 0.65	0.6336	(close)
Stability	< 0.02	0.0186	Good Threshold optimisation
> 0.70	0.6945	(close)	
Segment performance	> 0.60	0.61-0.66	Good
Overfitting	< 0.05	0.0014	Good

6.2 Suitability for the Use Case

The model is suitable for the HealthConnect use case because:

- It is stable (standard deviation < 0.02)
- There is no overfitting
- It performs across all segments
- The optimal threshold improves the F1 by +5.1 points

6.3 Limitations

Table 15: Model Limitations

Limitation	Description
F1 slightly below objectives	Gaps from -0.005 to -0.016
False Negatives	161 undetected cases
Fictional data	Limited generalisation

6.4 Recommendations for Week 8

1. **Optimise hyperparameters** to reach $F1 > 0.70$
2. **Reduce False Negatives** (161 cases)
3. **Integrate with the ML Engineering pipeline**
4. **Validate with stakeholders**
5. **Prepare the final presentation**

7 Conclusion

7.1 Summary of Achievements

This week made it possible to:

- Carry out 6 systematic tests
- Validate stability and absence of overfitting
- Optimise the decision threshold (+5.1 F1 points)
- Collaborate with Data Analytics
- Prepare the final validation for Week 8

7.2 Major Scientific Result

The candidate model is **stable**, **free of overfitting** and **homogeneous** across all segments. Although 3 out of 6 tests failed, the gaps are very small (between -0.005 and -0.016), confirming that the model is **ready for Week 8** with minor improvements.

7.3 Contribution to the HealthConnect Project

- Validation of the model's robustness
- Optimisation of the decision threshold
- Alignment with business KPIs
- Solid foundation for the final presentation

7.4 Week 8 Perspectives

- Hyperparameter optimisation
- Reduction of False Negatives
- Final integration with ML Engineering
- Presentation of results

Appendices

Appendix A: Detailed Results

Table 16: 10-fold Cross-Validation - Detail

Model	Average F1	Standard deviation	Status
Baseline (LR)	0.6448	0.0204	Close
Random Forest	0.6221	0.0247	Less good

Appendix B: Performance by Segment

Table 17: Detailed Performance by Segment

Segment	F1	Patients	Criterion	Status
18-34	0.6549	267	> 0.60	Good
35-49	0.6589	231	> 0.60	Good
50-64	0.6412	237	> 0.60	Good
65+	0.6108	213	> 0.60	Good

Appendix C: References

- AnalystLab Africa Experience Lab - Week 7 Assignment Document
- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base